KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2013
Department of Computer Science and Engineering
CSE 4215
Information Security and Control

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Define the terms: 'Ciphertext-only attack', 'Known-plaintext attack' and 'Chosen-plaintext attack'. Discuss Shannon characteristics of good cipher. (12)
   b) Explain signing and verification mechanisms of ElGamal Digital Signature scheme with an example. (13)
   c) Discuss how can a computer network become vulnerable? Briefly explain the reasons. (10)

2. a) Discuss the terms: 'Stream ciphers' and 'Block ciphers'. Discuss a specific key exchange protocol to distribute a key. (12)
   b) Discuss the characteristics of relation between the two keys of public key cryptography. Explain how RSA keys are generated employing extended Euclid's gcd algorithm. (13)
   c) Define sensitive data. Discuss the factors that can make data sensitive. (10)

3. a) Discuss the basic protocol of signing documents with public-key cryptography and one-way hash function. (08)
   b) ElGamal encryption mechanism is non-deterministic, how? Explain with a suitable example. (12)
   c) Explain the steps of AEs algorithm with a block diagram. (08)
   d) Discuss the threats to E-mail. Also discuss how to defend against it. (07)

4. a) Define cryptographic hash function. Discuss the properties of it. (08)
   b) Explain the mechanism of triple DES encryption. (07)
   c) Explain the homomorphic encryption mechanism with possible merits and demerits. (10)
   d) Discuss the dimensions of reliability and integrity of database. What are the requirements for database security? Explain them. (10)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) What is information security? "Security measures must balance between security and productivity"- justify. (12)
   b) Define risk and risk analysis. Discuss the steps of risk analysis. (11)
   c) Write notes on- i) Confidentiality, ii) Integrity, and iii) Availability (12)

6. a) Differentiate between cryptography and cryptanalysis. Encrypt the following message using column transposition method with five column and Vernam method. (15)
   "CAESAR WAS KILLED BY BRUTUS"
   b) What is the basic theme of OS security? What is fence register? How does fence register provide extra benefit in relocation method? Explain. (12)
   c) What are the differences between paging and segmentation? (08)

7. a) What are malicious error and non-malicious error? Discuss six categories of non-malicious errors. (10)
   b) Explain risk exposure and risk leverage with example. (10)
   c) Discuss exhaustive attack and one time password with an example. Differentiate between copyright, patents, and trade secret. (15)

8. a) Differentiate between laws and ethics. Why is computer crime hard to define and hard to prosecute? Explain. (10)
   b) Explain virus signature with an example. Differentiate between virus and worm. (10)
   c) Explain buffer overflow with an example. How does buffer overflow affect code and data? Discuss with necessary figures. (15)

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